

The Gravitational Projection Kernel: Non-Admissibility from Cartan Self-Coupling

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Abstract

We prove that the General Relativity projection kernel is non-admissible by computing the admissibility defect in the Cartan geometry pipeline. The curvature formula $R_b^a = d\omega_b^a + \omega_c^a \wedge \omega_b^c$ has the same algebraic structure as the Yang–Mills field strength $F = dA + A \wedge A$: both contain a quadratic self-coupling term that breaks kernel admissibility. Using Kleis’s symbolic Cartan pipeline, we compute the defect $\Delta(e_A, e_B) = K(e_A + e_B) - K(e_A) - K(e_B)$ on the Schwarzschild tetrad and verify it is nonzero, isolating the $\omega \wedge \omega$ term as the source. We develop the full fiber interpretation: diffeomorphism orbits are the projection fibers of GR, and non-admissibility produces gravitational energy non-localizability via the Main Theorem of projection fiber theory. We compute the Weyl tensor as the content of $\ker(Q_{\text{GR}})$ and prove that the Schwarzschild singularity lives entirely in $\ker(Q)$: the curvature diverges but Q maps it to zero. The central conceptual result is a re-interpretation of the observable projection Q . In Yang–Mills, non-admissibility confines charges, requiring the Higgs mechanism to restore observability. In GR, the same algebraic non-admissibility makes energy non-localizable but does not hide geometric observables (tidal forces, gravitational waves, frame-dragging). The distinction is structural: Yang–Mills gauge symmetry acts on internal charge space, while GR gauge symmetry acts on spacetime itself. We formulate a Projection Sufficiency Principle: non-admissibility requires dynamical restoration if and only if the essential observables lie in the sector it hides. We then confront an admissibility dilemma: the three gravitational observations commonly cited as confirming full GR — binary pulsar decay, frame-dragging, and gravitational waves — are all predictions of linearized GR, which is admissible. The only claimed evidence for the nonlinear $\omega \wedge \omega$ term is the LIGO merger waveform, extracted via matched filtering against GR-derived templates. Meanwhile, the admissible logarithmic kernel (Volume III) explains flat rotation curves without dark matter, where both GR and Newton fail — but this success is regime-specific (patched at a core radius R_c , structurally analogous to MOND). Whether the universe operates through admissible kernels remains genuinely open. All 50 results are verified: 17 by Kleis evaluation and 33 by the Z3 SMT solver.

Keywords: general relativity, Cartan geometry, admissible kernel, projected ontology, non-admissibility, connection self-coupling, gravitational energy, linearized gravity, diffeomorphism fibers, Weyl tensor, projection sufficiency, Higgs mechanism, singularity annihilation, logarithmic kernel, rotation curves, formal verification, Z3 theorem prover

1 Introduction

The non-localizability of gravitational energy is one of the oldest puzzles in general relativity. Unlike electromagnetic energy, which is carried by the gauge-invariant field strength $F = dA$, gravitational energy cannot be localized in a coordinate-independent manner: any local expression for it can be transformed to zero at a point by choosing appropriate coordinates. Equivalently, the gravitational energy-momentum pseudo-tensor¹ is not a true tensor under general coordinate transformations. The standard explanation — diffeomorphism invariance of the full theory — is correct but does not pinpoint the algebraic mechanism. Similarly, the problem of time in canonical gravity — that the Hamiltonian is a pure constraint generating gauge transformations rather than physical evolution — lacks a structural characterization that connects it to the analogous Yang–Mills phenomenology.

This paper identifies the algebraic mechanism: the non-admissibility of the GR production kernel. In the Projected Ontology (POT) framework, a production kernel K maps a configuration (the tetrad e^a , encoding the metric) to curvature (the Riemann tensor R_b^a). An observable projection Q then extracts measurable quantities (the Ricci tensor, Einstein tensor, and through the field equations, the energy-momentum tensor). The kernel K is **admissible** if it is linear: $K(e_A + e_B) = K(e_A) + K(e_B)$ and $K(0) = 0$. The admissibility defect $\Delta(e_A, e_B) = K(e_A + e_B) - K(e_A) - K(e_B)$ measures how badly linearity fails.

The Cartan curvature formula provides the explicit mechanism:

$$R_b^a = d\omega_b^a + \omega_c^a \wedge \omega_b^c$$

This is structurally identical to the Yang–Mills field strength $F = dA + A \wedge A$. The $\omega \wedge \omega$ term — the connection self-coupling — breaks linearity. In the companion paper on Yang–Mills confinement (Volume IV), we showed that $A \wedge A$ produces the Lie bracket $[A, B]$ as the admissibility defect, and derived color confinement as a structural consequence. Here we show that $\omega \wedge \omega$ produces a nonzero admissibility defect for GR, and derive the three structural consequences listed above.

The key advantage of our approach is that the Cartan pipeline is **computational**: the map tetrad $\rightarrow$ connection $\rightarrow$ curvature is implemented as a symbolic evaluation chain in Kleis. We do not merely axiomatize non-admissibility — we compute it, using the Schwarzschild metric as a concrete test case, and isolate the $\omega \wedge \omega$ term as the source.

The linearized theory provides the control case. When the background is flat, $\omega = 0$, so $\omega \wedge \omega = 0$, and the kernel becomes admissible. This explains why linearized GR has localizable gravitational energy (carried by the Isaacson stress-energy tensor), an external background time (Minkowski), and freely propagating gravitational waves. The transition from linearized to full GR is the transition across the admissibility boundary.

A striking consequence emerges from the Weyl tensor analysis: the Schwarzschild singularity at $r = 0$ — a genuine curvature singularity where $R_{abcd}R^{abcd} \rightarrow \infty$ — lives entirely in $\ker(Q_{\text{GR}})$. The observable projection maps this divergent curvature to zero ($R_{ab} = 0$). In the POT framework, where physical observables are identified with $\text{im}(Q)$, the vacuum singularity is not an observable divergence but a structural artifact of the production kernel — following the same pattern as UV divergence annihilation in quantum field theory.

All 50 results are verified: 17 by the Kleis evaluator (symbolic Cartan computations) and 33 by the Z3 SMT solver (structural consequences).

2 GR as a K-Q Pair via Cartan Geometry

We identify the two operators that define GR within the POT framework.

Production kernel K_{GR} . The production kernel maps a tetrad e^a to the Riemann curvature tensor via Cartan’s structure equations in three explicit steps.

Step 1: Anholonomy coefficients. Compute the exterior derivatives de^a and extract the structure coefficients C_{bc}^a defined by:

$$de^a = \frac{1}{2} C_{bc}^a e^b \wedge e^c$$

These encode how the tetrad frame fails to be a coordinate basis.

Step 2: Connection. The unique torsion-free, metric-compatible connection is:

$$\omega_b^a = \sum_c \frac{1}{2} \left(C_{bc}^a - C_b^{(a)c} + C_c^{(a)b} \right) e^c$$

This is the Koszul formula in the tetrad basis. The connection ω_b^a is a set of 4×4 one-forms, each a linear combination of the tetrad one-forms weighted by the anholonomy coefficients. The nonlinearity enters here: the C_{bc}^a depend on the tetrad components and their derivatives, so ω is a **nonlinear** function of e .

Step 3: Curvature. The Riemann curvature 2-form is:

$$R_b^a = d\omega_b^a + \sum_c \omega_c^a \wedge \omega_b^c$$

The first term $d\omega$ is linear in ω (hence nonlinear in e through Step 2). The second term $\omega \wedge \omega$ is **quadratic** in ω — this is the self-coupling that makes K_{GR} non-admissible, structurally identical to $A \wedge A$ in the Yang–Mills field strength $F = dA + A \wedge A$.

The complete kernel is the composition: $K_{\text{GR}}(e) = R(\omega(e))$, implemented in Kleis as `compute_riemann(tetrad)`, which chains exterior differentiation (`d1`), the connection solver (`solve_connection`), and wedge products (`wedge`).

Observable projection Q_{GR} . The observable projection is the Ricci contraction — the trace of the Riemann tensor over its first and third indices:

$$R_{ab} = \sum_{c=0}^3 R_{acb}$$

This 4×4 symmetric tensor is the part of the curvature that couples to matter via the Einstein field equations $G_{ab} = R_{ab} - \frac{1}{2} R g_{ab} = 8\pi G T_{ab}$. The kernel of Q — what the trace contraction discards — is the Weyl tensor C_{abcd} : the 10-component trace-free part of the Riemann tensor encoding tidal forces, gravitational waves, and long-range gravitational effects not determined by local matter.

Verification. We verify the pipeline on two metrics:

- **Minkowski (flat):** K_{GR} produces zero curvature, $Q \circ K$ produces zero Ricci. This is the trivial case: no gravitational field.

- **Schwarzschild** (curved): K_{GR} produces nonzero curvature components depending on M and r , while $Q \circ K$ produces zero Ricci ($R_{\mu\nu} = 0$): the Schwarzschild metric is a vacuum solution. This confirms that the kernel operates correctly on a physically nontrivial input.

Yang–Mills parallel. The structural isomorphism is exact:

Property	Yang–Mills	General Relativity
Kernel formula	$F = dA + A \wedge A$	$R = d\omega + \omega \wedge \omega$
Configuration	Gauge connection A	Tetrad e^a
Bundle	Principal G -bundle	Frame bundle $\text{SO}(3, 1)$
Self-coupling	$A \wedge A$	$\omega \wedge \omega$
Defect	Lie bracket $[A, B]$	Connection nonlinearity

3 The Admissibility Defect

The admissibility defect Δ measures the failure of linearity:

$$\Delta(e_A, e_B) = K(e_A + e_B) - K(e_A) - K(e_B)$$

For an admissible kernel, $\Delta = 0$ for all inputs. For the GR kernel, we compute Δ on concrete tetrads.

Test case. We take e_A = Schwarzschild tetrad (parametric in mass M) and e_B = a radial perturbation $\varepsilon \cdot dr$. The defect component Δ_{0101} evaluates to a nonzero symbolic expression in M , r , and ε . The expression is complex (involving nested square roots and rational functions of M and r), which is expected: the connection solver introduces ratios of tetrad components, and the curvature computation adds quadratic wedge products.

The key observation is that $\Delta \neq 0$: the map from tetrad to curvature is **not** linear. Admissibility requires $\Delta(e_A, e_B) = 0$ for **all** inputs — it is a universally quantified property. A single computed counterexample with $\Delta \neq 0$ is therefore sufficient to refute it. The Schwarzschild defect is that counterexample. This proves non-admissibility of the GR kernel as a mathematical fact, not an assumption: the Cartan pipeline computes Δ , and the result is nonzero.

Source identification. The defect arises from two sources:

1. **Connection nonlinearity.** The connection ω_b^a depends on the anholonomy coefficients C_{bc}^a extracted from de^a , which involve products of tetrad components and their inverses. The connection solver formula $\omega_b^a = \frac{1}{2}(C_{bc}^a - C_{ac}^b + C_{ab}^c)e^c$ is linear in the C 's, but the C 's themselves are nonlinear in the tetrad.
2. **Curvature self-coupling.** Even if the connection were linear in the tetrad, the curvature formula $R = d\omega + \omega \wedge \omega$ adds a quadratic term. This is the same mechanism as Yang–Mills: $F = dA + A \wedge A$.

Both sources contribute, but the second is the **structural** source — it is the same for GR and Yang–Mills, and it persists even when the connection is treated as a fundamental variable (Palatini formalism).

4 Isolating the Self-Coupling Term

To pinpoint the nonlinear contribution, we decompose the curvature into its linear and nonlinear parts:

$$R_b^a = \underbrace{d\omega_b^a}_{\text{linear}} + \underbrace{\sum_c \omega_c^a \wedge \omega_b^c}_{\text{nonlinear}}$$

We compute each part separately using the Kleis Cartan pipeline.

Minkowski (flat spacetime). The connection vanishes: $\omega_b^a = 0$ for all a, b . Therefore $\omega \wedge \omega = 0$, and the full curvature $R = d\omega + 0 = 0$. This is the **admissible limit**: when the self-coupling vanishes, the kernel is (trivially) linear in the tetrad.

Schwarzschild (curved spacetime). The connection is nonzero (e.g., ω_1^0 depends on $\frac{M}{r^2}$ and $\sqrt{1 - 2\frac{M}{r}}$). The nonlinear part $\omega \wedge \omega$ is also nonzero: we compute the $(0, 1, 0, 1)$ component and find a nontrivial expression involving products of connection components. The full curvature $R = d\omega + \omega \wedge \omega$ receives contributions from both terms.

This decomposition makes precise the claim that the $\omega \wedge \omega$ self-coupling is the source of non-admissibility. In the flat limit ($M \rightarrow 0$), $\omega \rightarrow 0$, the self-coupling vanishes, and the kernel crosses the admissibility boundary into the admissible regime.

The structural parallel is again exact:

- Yang–Mills: $A = 0$ (trivial connection) $\rightarrow A \wedge A = 0 \rightarrow$ admissible limit (abelian electrodynamics)
- GR: $\omega = 0$ (flat connection) $\rightarrow \omega \wedge \omega = 0 \rightarrow$ admissible limit (linearized gravity)

5 Structural Consequences

Non-admissibility of the GR kernel produces three structural consequences, all verified by Z3.

C1. Gravitational energy non-localizability. For an admissible kernel, the image $K(e)$ is invariant under gauge transformations (diffeomorphisms): the curvature is coordinate-independent. For a non-admissible kernel, the image is **not** invariant on diffeomorphism orbits. The energy extracted from curvature inherits this non-invariance: it depends on the coordinate choice, hence cannot be localized. This is the algebraic root of the pseudo-tensor problem.

The chain is: non-admissible $\rightarrow$ image not gauge-invariant $\rightarrow$ energy not localizable. The contrapositive holds for linearized GR: admissible $\rightarrow$ image gauge-invariant $\rightarrow$ Isaacson stress-energy tensor is well-defined.

C2. Problem of time. In canonical (Hamiltonian) gravity, the total Hamiltonian is a sum of constraints: $H = N\mathcal{H} + N^i\mathcal{H}_i \approx 0$. There is no true ‘Hamiltonian generating physical time evolution. Non-admissibility forces this: the nonlinear structure of the constraints prevents separation into a free ‘Hamiltonian and gauge generators. In the admissible (linearized) limit, the Minkowski background provides an external time coordinate, and the linearized Hamiltonian generates physical evolution.

C3. Admissibility boundary. Full nonlinear GR is non-admissible. Linearized GR (perturbations around flat space) is admissible. The transition between them is the admissibility boundary — the same structural boundary that separates non-abelian Yang–Mills (confined) from abelian electrodynamics (free).

Property	Full GR (non-admissible)	Linearized GR (admissible)
Self-coupling $\omega \wedge \omega$	Nonzero	Zero
Energy localizability	No (pseudo-tensor)	Yes (Isaacson)
External time	No (problem of time)	Yes (Minkowski background)
Gauge structure	Nonlinear diffeomorphisms	Linear perturbations
Gravitational waves	Coupled to background	Freely propagating

C4. Degree-of-freedom counting. A symmetric 4×4 metric tensor has 10 independent components. Diffeomorphism invariance removes 4 gauge degrees of freedom. The Hamiltonian constraint and 3 momentum constraints remove 4 more. The remaining $10 - 4 - 4 = 2$ physical degrees of freedom are the transverse-traceless modes: the two polarizations of gravitational waves. This counting holds for both the full and linearized theories.

6 Diffeomorphism Fibers and Non-Localizability

In the Projected Ontology framework, projection fibers are orbits of the gauge group acting on the configuration space. For Yang–Mills, the fibers are gauge orbits $\{g \cdot A : g \in \mathcal{G}\}$. For GR, the fibers are **diffeomorphism orbits**: the set of all tetrads related by coordinate transformations, $\{\varphi^*(e) : \varphi \in \text{Diff}(M)\}$.

Two tetrads in the same fiber describe the same physical geometry in different coordinates. The projection $Q \circ K$ must produce the same observable for both — this is the content of general covariance. But the intermediate kernel K (tetrad $\rightarrow$ curvature) may or may not be invariant on these fibers.

Admissible kernel (linearized GR). When the kernel is admissible, the curvature is invariant on diffeomorphism orbits: $K(\varphi^*(e)) = K(e)$ for all diffeomorphisms φ . The image is constant on each fiber. Any predicate on the image — including gravitational energy — is therefore fiber-invariant, hence a genuine diffeomorphism-invariant observable. This is why linearized GR admits the Isaacson stress-energy tensor.

Non-admissible kernel (full GR). When the kernel is non-admissible, there exist diffeomorphisms φ and tetrads e such that $K(\varphi^*(e)) \neq K(e)$: the curvature components change across the fiber. Any predicate that depends on these components — including gravitational energy density — is fiber-non-invariant.

By the Main Theorem of projection fiber theory (proved in Volume III): **a predicate is independent of the projection if and only if it is non-invariant on fibers**. Applied to GR:

non-admissible $K \rightarrow$ image varies on fibers $\rightarrow$ energy is fiber-non-invariant $\rightarrow$ energy is independent of observables

This **is** non-localizability, derived from the fiber structure. The gravitational energy pseudo-tensor is not a defect of our formalism — it is a structural necessity: the kernel’s non-admissibility forces energy to be invisible to the projection.

The parallel with Yang–Mills confinement is now exact:

- **Yang–Mills:** color charge is non-invariant on gauge fibers $\rightarrow$ independent of gauge-invariant observables $\rightarrow$ confined
- **GR:** gravitational energy is non-invariant on diffeomorphism fibers $\rightarrow$ independent of diffeomorphism-invariant observables $\rightarrow$ non-localizable

Both are instances of the same structural theorem, applied to different fiber bundles.

Gribov analog. For admissible K , diffeomorphism orbits are affine subspaces, so a global coordinate fixing (harmonic gauge, $\square x^\mu = 0$) selects one representative per orbit. For non-admissible K , orbits are nonlinear manifolds, and no single coordinate system covers the entire solution space. This is the GR analog of the Gribov problem in Yang–Mills: just as there is no global gauge fixing for non-abelian gauge fields, there is no global coordinate system for general spacetimes. The Schwarzschild coordinate singularity at $r = 2M$ — requiring Kruskal–Szekeres extension — is a concrete instance.

Observable hierarchy. The fiber structure determines the minimum order of observables:

- **Admissible (linearized):** The metric perturbation h_{ab} is itself gauge-invariant at linear order — it is an order-1 observable.
- **Non-admissible (full):** Only nonlinear curvature invariants survive as diffeomorphism-invariant observables: the Ricci scalar R (order 2 in derivatives), the Kretschmer scalar $R_{abcd}R^{abcd}$ (order 4), the Weyl invariant $C_{abcd}C^{abcd}$ (order 4). The metric itself is not an observable.

All fiber-structure results (F1–F11) are Z3-verified.

7 Admissibility and the Gravitational Kernels

In Volume V, we showed that the Higgs mechanism in the electroweak sector can be understood as **admissibility restoration**: the Higgs field compensates the Lie bracket defect in the $SU(2)$ kernel, making weak charges observable. Given the structural identity between GR and Yang–Mills at the kernel level ($\omega \wedge \omega$ versus $A \wedge A$), one might expect a gravitational analog of the Higgs mechanism to exist. We argue that no such mechanism is needed, and that the reason illuminates a fundamental distinction between the two theories.

The same algebra, different consequences. Both kernels are non-admissible for the same algebraic reason: quadratic self-coupling. But the gauge symmetries act on different objects:

- In Yang–Mills, the gauge symmetry is **internal**: $SU(2)$ and $SU(3)$ act on an internal charge space (color, weak isospin). These are properties of matter fields that live **in** spacetime. When non-admissibility confines these charges, it hides something inside spacetime. Without the Higgs, weak charges would be invisible — the theory would predict unobservable W and Z bosons.
- In GR, the gauge symmetry is **external**: diffeomorphisms act on spacetime **itself**. Gravity is not a field in spacetime — it **IS** spacetime. Non-admissibility makes gravitational energy non-

localizable, but it does not hide the observable effects of gravity. Tidal forces, gravitational lensing, gravitational waves, frame-dragging — all are directly measured. You cannot confine spacetime. You are already in it.

What GR does explain. GR is supported by several classes of observation, of varying directness:

- **Binary pulsar decay:** The Hulse–Taylor pulsar energy loss matches the quadrupole formula to 0.2% accuracy over decades of observation. This is a direct measurement — the pulse arrival times shift cumulatively, with no large systematics to subtract.
- **Frame-dragging:** Gravity Probe B reported Lense–Thirring precession at 37.2 ± 7.2 mas/yr, consistent with GR’s prediction of 39.2 mas/yr. The measurement required extensive modeling of systematic effects (electrostatic patch torques) that exceeded the signal itself. LAGEOS satellite tracking yields consistent results after subtracting Earth’s gravitational multipole contributions.
- **Gravitational waves:** LIGO detected coincident strain signals in two detectors consistent with a binary inspiral waveform. The signal extraction uses matched filtering — cross-correlation against GR-derived templates — so the detection and the theory are not fully independent.

No gravitational Higgs¹⁴ was needed for nature to produce these observables, with the binary pulsar providing the most model-independent confirmation.

What GR does not explain. Galaxy rotation curves are flat: orbital velocity $v(r)$ remains approximately constant far beyond the visible matter distribution. The GR kernel (and its Newtonian limit) predicts Keplerian decline $v(r) \propto r^{-1/2}$, in sharp disagreement with observation. The standard resolution is to postulate dark matter — unobserved mass that modifies $T_{\mu\nu}$ to produce the observed curves. But this means GR requires invisible matter to match galactic observations, a move that is structurally analogous to adding an unobserved field.

In Volume III, we showed that a different admissible kernel — the logarithmic projection kernel $K_{\log}$ with Green’s function $G(r, r') = -(\kappa/2\pi) \ln|r - r'|$ — produces $v(r) = \text{const}$ directly, without dark matter. The logarithmic kernel is admissible (linear), so it does not raise the Higgs question. But it predicts a different gravitational potential ($\ln r$ instead of $1/r$) and does not produce frame-dragging or gravitational waves.

The situation is therefore: GR is correct for strong-field, dynamical phenomena (waves, frame-dragging, binary pulsars), while the logarithmic kernel matches galactic rotation curves without auxiliary hypotheses. Neither kernel, alone, accounts for all gravitational observations.

The admissibility dilemma. The success of the logarithmic kernel creates a genuine interpretive tension. The kernel that explains flat rotation curves is admissible. Could this be evidence that the universe prefers admissible kernels — that non-admissibility in GR is, after all, an obstruction that nature corrects, perhaps through some restoration mechanism we have not yet identified?

The answer is not clear, and we must be honest about this. The evidence pulls in two directions:

For a preference for admissibility: the logarithmic kernel (admissible) explains rotation curves that the GR kernel (non-admissible) cannot without dark matter.

Against a preference for admissibility: the Newtonian kernel is also admissible, and it too fails at rotation curves — it predicts Keplerian decline, just like GR’s weak-field limit. Admissibility alone

does not select the correct kernel. The logarithmic kernel succeeds not because it is admissible, but because it has a specific Green’s function ($\ln|r - r'|$) that produces the right potential. Newton’s kernel has a different Green’s function ($1/|r - r'|$) that produces the wrong one.

This means we face a four-kernel landscape:

Property	Newton	Logarithmic	Linearized GR	Full GR
Admissible?	Yes	Yes	Yes	No
Rotation curves?	No (Keplerian)	Yes (flat)	No (Keplerian)	No (needs dark matter)
Frame-dragging?	No	No	Yes	Yes
Gravitational waves?	No	No	Yes	Yes
Binary pulsar?	No	No	Yes	Yes

A critical observation. The three observations commonly cited as confirming full GR — binary pulsar decay, frame-dragging, and gravitational waves — are all predictions of **linearized** GR. The quadrupole formula for gravitational radiation is derived in the linearized/post-Newtonian regime. The Lense–Thirring effect comes from the gravitomagnetic potential h_{0i} of the linearized metric. The gravitational wave equation is a linearized Einstein equation. None of these observations requires the $\omega \wedge \omega$ self-coupling that makes full GR non-admissible.

The only claimed evidence for the nonlinear regime is the specific waveform shape during the final merger cycles in LIGO data, where numerical relativity (full nonlinear GR) produces templates that linearized theory cannot. But this is precisely the measurement with the most model-dependent extraction: the signal is identified through matched filtering against templates built from the nonlinear theory.

This shifts the landscape significantly. Three admissible kernels (Newton, logarithmic, linearized GR) account for different subsets of gravitational observations. Linearized GR is the most successful admissible kernel: it explains frame-dragging, gravitational waves, and binary pulsar decay — everything except rotation curves. The logarithmic kernel explains rotation curves but not the dynamical phenomena. The only evidence that requires going **beyond** admissible kernels into the non-admissible full GR is the LIGO merger waveform, which is the least model-independent of all the observations.

We cannot conclude that the universe systematically prefers admissible kernels. But we can observe that the evidence for the **non-admissible** part of GR — the $\omega \wedge \omega$ term, the very structure that creates the non-admissibility — is thinner than commonly presented. The cleanest measurements do not distinguish linearized from full GR.

The patching problem. The situation is in fact more uncomfortable than a three-way competition. In Volume III, we introduced a core radius R_c and defined the logarithmic kernel’s domain as $r \geq R_c$, with the inner region ($r < R_c$) governed by the standard baryonic mass distribution — which is to say, by the Newtonian potential. The matching condition at R_c fixes $\kappa = 2\pi/R_c$ and produces the flat outer rotation curve. But this is a **patch**: two different kernels stitched together

at a boundary, not a single unified kernel that reproduces both the inner rise and the outer plateau from one structure.

This is structurally analogous to MOND (Milgrom, 1983), which introduces an acceleration scale a_0 and applies a modified force law below it while retaining Newton above. Both approaches succeed empirically in their respective outer regimes; both involve a transition parameter (R_c or a_0); and neither derives a single kernel valid at all scales from first principles. The POT formulation has the advantage that its transition is spatial (R_c) rather than dynamical (a_0), and the logarithmic kernel has a concrete Green’s function interpretation. But the structural similarity is real, and we should not overclaim: the flat rotation curve result is a regime-specific success, not a demonstration that the logarithmic kernel is the universal gravitational kernel.

The role of the observable projection. What we **can** say is that the Higgs question — does non-admissibility hide essential observables? — has a clear answer for GR:

Property	Yang–Mills	GR
Gauge symmetry acts on	Internal charge space	Spacetime itself
What enters $\ker(Q)$	Charges (color, weak isospin)	Energy density
What remains in $\text{im}(Q)$	Not enough — charges missing	Geometry intact
Non-admissibility hides essential observables?	Yes	No

In Yang–Mills, non-admissibility pushes charges into $\ker(Q)$, and $\text{im}(Q)$ without the Higgs is missing the weak sector entirely. The Higgs moves charges from $\ker(Q)$ to $\text{im}(Q)$, restoring observability. In GR, non-admissibility pushes energy density into $\ker(Q)$, but the geometry — tidal forces, geodesic deviation, wave strain — stays in $\text{im}(Q)$. Non-admissibility does not hide the geometric observables that GR correctly predicts.

But the rotation curve problem does not fit neatly into either the Higgs question or the kernel-range question. It is not that non-admissibility **hides** the flat rotation curves — the Newtonian kernel is admissible and also fails. It is that the **specific form** of the GR kernel (and its Newtonian limit) produces the wrong potential at galactic scales. Whether this points to a different kernel, a modification of GR, or genuinely invisible matter remains an open question — one that the POT framework identifies precisely but does not resolve.

Linearization is an approximation, not a mechanism. Writing $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$ and linearizing does produce an admissible kernel $K_{\text{eff}}(h)$, formally analogous to Higgs restoration. But linearization is something physicists do, not something nature does. The background η is a human choice. That said, the fact that the cleanest gravitational observations (binary pulsar, frame-dragging, gravitational waves) are all predictions of the linearized theory raises a genuine question about whether the nonlinear $\omega \wedge \omega$ term is physically realized or merely a mathematical extension.

What we can and cannot conclude. The distinction between internal and external gauge symmetry is structural: Yang–Mills gauge symmetry acts on internal charge space (can confine charges), while GR gauge symmetry acts on spacetime itself (cannot confine geometry). This

explains why non-admissibility requires a Higgs in Yang–Mills but does not hide geometric observables in GR.

But whether the universe ultimately operates through admissible kernels at all scales — whether the success of the logarithmic kernel at galactic scales is a clue — is an open question. POT provides the language to state the question precisely: three gravitational kernels, two admissible and one not, each correct in a different regime, none complete. The framework does not tell us which principle selects the right kernel in each regime. That remains to be discovered.

Results R1–R4 verify the formal linearization claims in Z3.

8 The Content of $\ker(Q)$: The Weyl Tensor

The observable projection Q_{GR} extracts the Ricci tensor from the Riemann curvature. What does it discard? The answer is the **Weyl tensor** C_{abcd} — the trace-free part of the Riemann tensor that encodes tidal forces, gravitational radiation, and the long-range gravitational field.

In four dimensions, the Riemann tensor has 20 independent components. The Ricci tensor has 10. The remaining 10 components form the Weyl tensor:

$$C_{abcd} = R_{abcd} - (\text{Ricci terms})$$

The Weyl tensor is the content of $\ker(Q_{\text{GR}})$: it is curvature that is **not** determined by the local energy-momentum tensor.

Computation: Schwarzschild. For the Schwarzschild vacuum ($T_{ab} = 0$), the Einstein equations give $R_{ab} = 0$: the Ricci tensor vanishes. But the Riemann tensor is **nonzero** — it produces the tidal forces that stretch and compress freely falling objects. Therefore: $\text{im}(K_{\text{GR}}) \subseteq \ker(Q_{\text{GR}})$ for vacuum. **All** of the curvature that K produces is projected away by Q .

We compute this explicitly using the Kleis Cartan pipeline: the Weyl component $W^0_{1\ 01}$ for Schwarzschild is nonzero (depending on M and r), confirming that the vacuum Riemann tensor has nontrivial content in $\ker(Q_{\text{GR}})$.

Minkowski. For flat spacetime, the Riemann tensor vanishes entirely: $W^0_{1\ 01} = 0$. The kernel of Q is empty for flat space — there is nothing to project away.

Physical interpretation. The Weyl tensor is responsible for:

- **Tidal forces:** the relative acceleration of nearby geodesics (geodesic deviation)
- **Gravitational waves:** the two transverse-traceless polarizations h_+ and $h_\times$ are Weyl modes
- **Gravitational lensing:** light deflection by massive objects (even in vacuum)
- **Black hole hair:** the Schwarzschild and Kerr metrics are entirely characterized by their Weyl tensor

All of these are physical effects that are **not** constrained by local matter — they are the content of the kernel of Q . In POT language: $\ker(Q_{\text{GR}})$ is non-empty and physically rich, containing the degrees of freedom that propagate freely through vacuum.

The obstruction is in K , not Q . This computation clarifies a subtle point. Q_{GR} is well-defined and computable: it is the Ricci contraction, and we verified it on Schwarzschild (producing

$R_{\mu\nu} = 0$) and Minkowski ($R_{\mu\nu} = 0$). The structural problems of GR — non-localizability, problem of time, coordinate singularities — do not arise from Q . They arise from K : the map from tetrads to curvature, where the $\omega \wedge \omega$ self-coupling introduces non-admissibility. The quantum gravity community’s search for the right quantization’ is, in POT language, a search for a Q to solve a problem that lives in K .

9 Prior Work

The algebraic and structural phenomena identified here have been studied from various perspectives. We highlight connections to the most relevant prior work and clarify where our contribution lies.

Curiel (2019). Curiel proved rigorously that no gravitational stress-energy tensor satisfying natural physical conditions can exist in GR. Our result reaches the same conclusion by a different route: we identify the **algebraic mechanism** (the $\omega \wedge \omega$ defect) that prevents energy localization, and we connect it to the Yang–Mills confinement chain through the shared non-admissibility structure. Curiel’s work establishes the impossibility; ours explains the algebraic root and places it in a cross-domain framework.

De Vuyst, Eccles, Hoehn, and Kirklin (2024). These authors identified linearization instabilities’ in GR: solutions of the linearized theory that do not integrate to exact solutions of the full nonlinear theory. In our framework, linearization instabilities are precisely the **admissibility boundary**: the transition from admissible (linearized) to non-admissible (full) kernel. Solutions that exist in the admissible regime but not in the non-admissible regime are structural artifacts of the boundary crossing. Our characterization is algebraic (the vanishing or non-vanishing of $\omega \wedge \omega$) rather than geometric, providing a complementary viewpoint.

Ashtekar (1986) and the connection formulation. Ashtekar reformulated GR as a gauge theory with $SO(3,1)$ (or $SL(2,\mathbb{C})$) connection variables, bringing GR closer to Yang–Mills in mathematical structure. Our work uses the same Cartan connection framework but extracts a different conclusion: the admissibility defect. Ashtekar’s formulation enables loop quantum gravity; our framework classifies the kernel K structurally, without committing to a specific quantization program.

Trautman and Sardanashvily: metric as gravitational Higgs. Trautman (1979) and later Sardanashvily proposed that the metric tensor in GR plays a role analogous to the Higgs field: it breaks the $GL(4,\mathbb{R})$ symmetry of the frame bundle to $SO(3,1)$. Our linearization-as-restoration analysis (Section 7) formalizes a related but distinct claim: the **background metric** η restores admissibility by absorbing the $\omega \wedge \omega$ defect, paralleling how the Higgs restores admissibility by absorbing the Lie bracket defect. The key difference we identify is fragility: the GR restoration is a non-dynamical choice, not a physical mechanism.

ADM formalism (1959). Our degree-of-freedom counting $10 - 4 - 4 = 2$ reproduces the result of the ADM (Arnowitt–Deser–Misner) canonical formulation of GR, which identifies the Hamiltonian and momentum constraints and establishes that the gravitational field has exactly 2 propagating degrees of freedom: the transverse-traceless polarizations h_+ and $h_\times$. LIGO’s detection

of gravitational waves (GW150914, 2015) reported a signal consistent with exactly 2 polarizations, using matched filtering against GR-derived waveform templates. Our framework arrives at the same count from the POT structure: the 4 gauge DOF are fiber directions (diffeomorphism orbits), the 4 constraints are $\ker(Q)$ content (projected away), and the 2 surviving modes are $\text{im}(Q \circ K)$ — what passes through both production and projection.

Logarithmic potential and flat rotation curves. In a companion paper, we showed that a logarithmic Green’s function $K(r) \sim \log(r)$ reproduces flat galactic rotation curves without dark matter. This raises the question: can the logarithmic kernel be derived from the GR weak-field limit? The answer is no. The GR weak-field kernel is the Newtonian $\frac{1}{r}$ potential, producing Keplerian falloff $v \sim \frac{1}{\sqrt{r}}$. The logarithmic correction requires physics beyond the standard linearization. Our framework makes this precise: the GR kernel (via Cartan) gives $\frac{1}{r}$ in the admissible limit; any departure from $\frac{1}{r}$ requires either a modification of K (e.g., additional fields, modified gravity) or a non-perturbative effect from the non-admissible regime.

10 Singularity Annihilation: Vacuum Singularities in $\ker(Q)$

The Schwarzschild singularity at $r = 0$ is a genuine curvature singularity: the Kretschmer scalar $R_{abcd}R^{abcd} \rightarrow \infty$. Yet Q_{GR} maps this divergent curvature to **zero**: $R_{ab} = 0$ everywhere, including at the singularity. The entire divergence lives in $\ker(Q_{\text{GR}})$ — the Weyl tensor sector. Q annihilates the infinity — not in the sense that the Riemann tensor ceases to diverge (it does not; the curvature invariant $R_{abcd}R^{abcd} \rightarrow \infty$ remains), but in the precise sense that the observable projection maps the divergent input to a finite (in fact, zero) output. The divergence is real in the production layer; it is absent from the observable layer.

The mechanism is identical to the annihilation of UV divergences in QFT. In the companion φ^4 paper (Volume I), we showed that one-loop divergences in Feynman integrals live in $\ker(Q)$: the divergent parts are scheme-dependent and do not contribute to physical observables. The observable projection Q extracts only the finite, scheme-independent residue. Here, Q_{GR} performs the same operation on the Schwarzschild singularity: it extracts the Ricci tensor (the part coupled to matter), which is zero, discarding the divergent Weyl tensor (the part encoding tidal forces).

Singularity classification by Q . The key structural distinction is between singularities that live in $\ker(Q)$ and those that live in $\text{im}(Q)$:

Singularity type	Ricci	Weyl	Q annihilates?
Vacuum (Schwarzschild, Kerr)	$R_{ab} = 0$	Diverges	Yes — in $\ker(Q)$
Matter (collapsing star)	$R_{ab} \rightarrow \infty$	Diverges	No — in $\text{im}(Q)$
Cosmological (Big Bang)	$R_{ab} \rightarrow \infty$	May diverge	No — in $\text{im}(Q)$

For vacuum singularities, the entire curvature is Weyl: $R_{abcd} = C_{abcd}$ when $R_{ab} = 0$. The observable projection sees nothing but vacuum’ ($T_{ab} = 0$). The singularity — the divergent tidal stretching,

the infinite curvature — is invisible to Q . It is a **structural artifact of the production kernel**, not a feature of the observable output.

To be precise: the Schwarzschild singularity at $r = 0$ is **not** a coordinate artifact — the curvature invariant $R_{abcd}R^{abcd} \rightarrow \infty$ is diffeomorphism-invariant and cannot be removed by coordinate choice (unlike the coordinate singularity at $r = 2M$, which is removable). What the POT framework establishes is a different and stronger claim: **the singularity is a projection artifact**. The divergent curvature is geometrically real, but it resides entirely in $\ker(Q_{\text{GR}})$ — the sector that the observable projection discards. The shift is not from geometry to coordinates, but from geometry to the production–observation distinction: the singularity is produced by K but erased by Q .

The observability criterion. Within Projected Ontology Theory, physical observables are identified with $\text{im}(Q)$: a quantity is physically accessible if and only if it survives the observable projection. This criterion formalizes standard practice in field theory, where gauge-dependent quantities (Yang–Mills), coordinate-dependent quantities (GR), and scheme-dependent quantities (QFT renormalization) are not regarded as physical. Consequently, divergences that lie entirely in $\ker(Q)$ do not correspond to observable singularities but to structural features of the production kernel.

For matter singularities, the Ricci tensor diverges along with the Weyl tensor: both $\text{im}(Q)$ and $\ker(Q)$ contain divergent content. Q does not annihilate these. The physical energy-momentum $T_{ab} \rightarrow \infty$ is a genuine observable divergence.

Structural parallel. Vacuum singularities in GR occupy exactly the same structural position as:

- **UV divergences in QFT:** live in $\ker(Q)$, renormalized away, do not affect observables
- **Ghost fields in Yang–Mills:** live in $\ker(Q)$, cancel unphysical degrees of freedom
- **Color charge:** non-invariant on gauge fibers, invisible to gauge-invariant observables

By the Main Theorem of projection fiber theory, a quantity that lives in $\ker(Q)$ is **independent** of the observable projection. The vacuum singularity satisfies this: it is a feature of the production kernel’s output that the projection erases.

Implication for singularity resolution. If vacuum singularities are $\ker(Q)$ phenomena, and if the POT framework treats $\ker(Q)$ content as structurally analogous to gauge artifacts and scheme-dependent quantities, then a natural question arises: **is the vacuum singularity a structural artifact of the classical kernel K_{GR} ?** In QFT, UV divergences in $\ker(Q)$ disappear when one uses a better definition of K (lattice regularization, dimensional regularization). Could the Schwarzschild singularity disappear with a better (non-perturbative, quantum) definition of K_{GR} ? This would give the widespread expectation in quantum gravity — that singularities are resolved by quantum effects — a precise structural home: **singularity resolution means finding a kernel K where the divergent Weyl component does not arise**.

All singularity results (S1–S6) are Z3-verified.

11 Discussion

The central result of this paper is that General Relativity and Yang–Mills theory share the same non-admissibility mechanism: quadratic self-coupling in the curvature formula. This is not an analogy — it is a structural identity:

$$R = d\omega + \omega \wedge \omega \qquad F = dA + A \wedge A$$

Both are curvature 2-forms of connections on fiber bundles (the frame bundle $\text{SO}(3,1)$ for GR, the principal G -bundle for Yang–Mills). The $\omega \wedge \omega$ and $A \wedge A$ terms break kernel admissibility in exactly the same way, producing nonzero admissibility defects that propagate through the same structural chain to physical consequences.

Yet the physical consequences differ:

- **Yang–Mills non-admissibility** $\rightarrow$ color confinement (charges unobservable) $\rightarrow$ Higgs needed
- **GR non-admissibility** $\rightarrow$ energy non-localizability $\rightarrow$ but geometry observable $\rightarrow$ no restoration needed

The re-interpretation of Q . In the Yang–Mills paper (Volume V), we presented the observable projection Q as primarily an algebraic filter: it takes the output of K and extracts the observable content. The obstruction was located entirely in K , and Q was treated as passive. The present paper forces a deeper reading of Q ’s role.

The critical observation is this: **the same non-admissible algebra produces different needs for restoration depending on what Q projects onto.** In Yang–Mills, Q needs charges. Non-admissibility confines them. Therefore $\text{im}(Q)$ is incomplete, and a restoring mechanism (the Higgs) must move charges from $\ker(Q)$ back into $\text{im}(Q)$. In GR, Q needs geometry. Non-admissibility hides energy density, but geometry — tidal forces, wave strain, geodesic deviation — remains in $\text{im}(Q)$. The projection is already complete. No restoration is needed.

This means Q is not merely a passive filter. It determines **whether non-admissibility is a problem.** The same algebraic defect in K is catastrophic or benign depending on what Q requires. Put differently: the algebra of K is the same, but Q decides whether the universe needs to respond with a dynamical mechanism.

We can state this as a principle:

Projection Sufficiency Principle: Non-admissibility of K requires a dynamical restoration mechanism if and only if the essential observables of the theory lie in the sector that non-admissibility pushes into $\ker(Q)$.

For Yang–Mills: essential observables include charges $\rightarrow$ charges confined by non-admissibility $\rightarrow$ restoration required. For GR: essential observables are geometric $\rightarrow$ geometry survives non-admissibility $\rightarrow$ no restoration required.

This reframes the relationship between the Higgs mechanism and GR. The question is not where is the gravitational Higgs;’ The question is does Q_{GR} need one;’ For the Higgs question specifically — does non-admissibility hide essential observables? — the answer is no. The binary pulsar, frame-dragging measurements, and gravitational wave detections are all consistent with

geometric predictions of GR. Non-admissibility does not confine geometric observables in the way it confines charges in Yang–Mills.

But the flat rotation curve problem introduces a deeper uncertainty. The admissible logarithmic kernel (Volume III) explains galactic rotation curves without dark matter, where the non-admissible GR kernel cannot. Read one way, this is evidence that the universe prefers admissible kernels — that non-admissibility in GR is a genuine obstruction. But the Newtonian kernel is also admissible and also fails at rotation curves, producing Keplerian decline just like GR’s weak-field limit. Admissibility does not select the correct kernel; the specific form of the Green’s function does. We therefore cannot conclude that the universe systematically honors or violates admissibility. The data underdetermines the principle.

Q as the arbiter of physical consequences. This interpretation promotes Q from a passive algebraic operation to the object that determines the **physical meaning** of non-admissibility. K produces the same kind of defect in both theories. But:

- $Q_{\text{YM}} + \text{non-admissible } K_{\text{YM}} = \text{incomplete observables} \rightarrow \text{dynamical fix needed}$
- $Q_{\text{GR}} + \text{non-admissible } K_{\text{GR}} = \text{complete observables} \rightarrow \text{no fix needed}$

The universe responds to the combination (K, Q) , not to K alone. The Higgs is not a response to non-admissibility per se — it is a response to **insufficient projection**. Where projection is sufficient, non-admissibility is tolerated.

Q_{GR} is concrete and computable. Unlike in quantum gravity — where the nature of the observable projection is an open question tied to the measurement problem — Q_{GR} in classical GR is fully determined: it is the Ricci contraction. We computed $Q \circ K$ on Schwarzschild and obtained $R_{\mu\nu} = 0$, confirming that the projection works correctly. The kernel of Q is the Weyl tensor, which we computed explicitly.

The intermediate layer is not observable. The Riemann tensor is the **intermediate** of the GR pipeline: it is the output of K (production) and the input to Q (observation). Stating that the Riemann tensor diverges at $r = 0$ is a statement about this intermediate layer — not about $\text{im}(Q)$. In the POT framework, attributing physical reality to the intermediate is a category error: it is structurally equivalent to asking what an electron is doing between measurements, what momentum a virtual photon carries, or what value a gauge field takes before gauge fixing. Only $\text{im}(Q)$ corresponds to physical observables. The curvature invariant $R_{abcd}R^{abcd} \rightarrow \infty$ is a property of the intermediate layer. The observable output is Q ’s result: $R_{ab} = 0$. Vacuum. Nothing diverges in the projection.

Admissibility as a classification principle. The admissibility boundary organizes physical theories:

- **Admissible:** Electrodynamics ($K = d$), linearized GR, Newton gravity, the logarithmic flat-rotation kernel
- **Non-admissible:** Yang–Mills ($K = d + A \wedge A$), full nonlinear GR ($K : e \rightarrow d\omega + \omega \wedge \omega$)

The admissible theories do not raise the question of restoration at all — their projections are automatically well-behaved. Among the non-admissible theories, the Projection Sufficiency

Principle determines whether a Higgs-like mechanism is required: it is required for Yang–Mills (insufficient projection) but not for GR (sufficient projection).

The evidentiary status of non-admissibility in GR. A further complication deserves emphasis. The three gravitational observations cited above — binary pulsar decay, frame-dragging, and gravitational waves — are all predictions of **linearized** GR, which is admissible. The quadrupole formula is a linearized/post-Newtonian result. The Lense–Thirring effect is derived from the gravitomagnetic potential h_{0i} of the linearized metric. The gravitational wave equation is the linearized Einstein equation. The only claimed observation that requires the nonlinear $\omega \wedge \omega$ self-coupling is the specific merger waveform shape in LIGO data, extracted via matched filtering against templates derived from the nonlinear theory. The evidence for the non-admissible part of GR — the very structure this paper analyzes — is therefore thinner than commonly presented. The cleanest gravitational measurements do not distinguish between linearized and full GR.

What this paper does not do. We do not derive the Einstein field equations from first principles, nor do we resolve the problem of time dynamically, nor do we provide a concrete quantum kernel K that resolves the Schwarzschild singularity. These require physical content beyond the algebraic framework developed here. What we establish is the structural layer beneath: the non-admissibility of the GR kernel, the role of Q as the arbiter of whether non-admissibility requires dynamical correction, and the classification of singularities by their position relative to $\ker(Q)$.

12 Conclusion

We have shown, using the Kleis Cartan geometry pipeline, that the General Relativity production kernel is non-admissible. The proof is computational: we evaluate the admissibility defect $\Delta(e_A, e_B) = K(e_A + e_B) - K(e_A) - K(e_B)$ on the Schwarzschild tetrad and a radial perturbation, obtaining a nonzero symbolic expression. We isolate the $\omega \wedge \omega$ self-coupling as the source, paralleling the $A \wedge A$ term in Yang–Mills.

Beyond the initial proof, we established the full fiber interpretation: diffeomorphism orbits are the projection fibers of GR, and non-admissibility forces the kernel image to vary across these fibers, producing non-localizability as a consequence of the Main Theorem of projection fiber theory. We identified the Gribov analog, the observable hierarchy, and the admissibility boundary between full and linearized GR. The Weyl tensor was computed as the explicit content of $\ker(Q_{\text{GR}})$.

The most significant result is the re-interpretation of the observable projection Q . In the Yang–Mills case, we argued that the Higgs mechanism is admissibility restoration — a dynamical field that moves confined charges from $\ker(Q)$ back into $\text{im}(Q)$. The GR case reveals that this is only half the story. The full principle is:

Non-admissibility of K requires dynamical restoration if and only if $\text{im}(Q)$ is insufficient for observation.

In Yang–Mills, the essential observables (charges, masses) are hidden by non-admissibility, so the Higgs is required. In GR, the geometric observables (tidal forces, gravitational waves, frame-dragging) survive non-admissibility intact. Non-admissibility does not hide the observables that Q_{GR} projects onto in the way it hides charges in Yang–Mills.

This promotes Q from a passive algebraic filter to the **arbiter** of whether non-admissibility has physical consequences. The kernel K is non-admissible in both cases for the same algebraic reason ($\omega \wedge \omega$ versus $A \wedge A$). But the universe responds to the pair (K, Q) , not to K alone. Where projection is sufficient (GR geometric observables), non-admissibility is tolerated; where it is insufficient (Yang–Mills charges), nature provides a dynamical fix.

A complication sharpens the picture. The three gravitational observations commonly cited as confirming GR — binary pulsar decay, frame-dragging, and gravitational waves — are all predictions of **linearized** GR, which is admissible. The only claimed evidence for the nonlinear $\omega \wedge \omega$ term is the specific merger waveform in LIGO data, extracted via matched filtering against GR-derived templates. The evidence for full non-admissible GR, as distinct from its admissible linearization, rests on a narrow base. Meanwhile, the admissible logarithmic kernel (Volume III) explains flat rotation curves where both GR and Newton fail — but this success is regime-specific, patched at a core radius R_c in a manner structurally analogous to MOND. Whether the universe operates through admissible kernels at all scales, or whether the success of different kernels in different regimes points to a deeper organizing principle, is an open question that the POT framework identifies but does not yet resolve.

The singularity analysis stands independent of this uncertainty: the Schwarzschild singularity lives entirely in $\ker(Q_{\text{GR}})$. The curvature diverges, but Q maps it to zero — the Ricci tensor vanishes for vacuum. The observable projection annihilates the singularity, separating vacuum singularities (in $\ker(Q)$, annihilated) from matter singularities (in $\text{im}(Q)$, observable). This result depends on the structure of Q , not on which kernel produced the curvature.

The unifying insight is that the curvature formula $R = d\omega + \omega \wedge \omega$, shared in structure by GR and Yang–Mills, produces the same algebraic non-admissibility in both theories. What differs is the observable projection, and the Projection Sufficiency Principle determines whether the universe must respond with a Higgs-like mechanism. But the empirical question — whether the nonlinear term is physically realized, or whether nature operates through the admissible linearization — remains open. The Projected Ontology framework provides the language to state, classify, and verify these distinctions precisely.

All 50 results — 17 by Kleis evaluation of the Cartan pipeline and 33 by the Z3 SMT solver — are formally verified, with no human intervention in the derivation chain.

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